COVID-19 Patient Risk Analysis

A tabular adaptation of a CT-based triage research question | UE24CS352A

Abstract

This project explores whether demographic and recorded clinical variables can rank COVID-19 records by ICU admission, intubation, and recorded death. It adapts the outcomes studied by Zhan and Li using the public Mexican surveillance CSV included in the repository. It does not reproduce their CT/radiomics method. The source has 1,048,575 rows, 21 columns, and 812,049 exact duplicate rows, with no patient identifier. Historical random-row metrics may therefore be optimistic and do not establish patient-level generalization.

Targets and features

ICU and intubation use known codes 1 (positive) and 2 (negative); other codes are unknown and excluded. All known labels for these outcomes occur among hospitalized cases (PATIENT_TYPE=2). Death is 1 for a valid dd/mm/yyyy date and 0 for 9999-99-99; malformed values are excluded. Feature groups are Demo (age, sex), DemoComorb (plus ten comorbidity flags), and Clinical (plus pneumonia). Outcome fields, PATIENT_TYPE, pregnancy, classification, and service/unit variables are excluded.

Data preparation and models

Age sentinels 97/98/99, comorbidity sentinels 98/99, and pneumonia 99 become missing. Training-fitted median/mode imputation, age scaling, and categorical one-hot encoding stay inside the pipeline. New splits group identical predictor rows while stratifying outcomes; this prevents exact feature-vector copies crossing partitions but cannot establish patient identity. Models are Logistic Regression, Random Forest, Linear SVM, MLP, and LightGBM. Independent methods include raw preprocessing, PCA (95% variance), L1 Logistic Regression selection, capped SMOTEENN, mutual-information SelectKBest (a GUS-inspired proxy), and ANOVA F-test SelectFpr. SMOTEENN resamples training data only and caps it at 100,000 rows by default.

Observed saved results

Outcome	Highest saved F1 configuration	Prec.	Recall	F1	ROC-AUC	AP
ICU	raw / Clinical / LightGBM	0.127	0.803	0.220	0.676	0.162
INTUBED	raw / Clinical / LightGBM	0.238	0.678	0.353	0.643	0.250
DEATH	smoteenn / Clinical / LightGBM	0.396	0.713	0.509	0.915	0.459

Target cohort counts

Target cohort	Known rows	Positive rows	Prevalence
ICU	192,543	16,858	8.75%
Intubation	192,706	33,656	17.46%
Recorded death	1,048,575	76,942	7.34%

These rows use historical stratified random row splits and default class thresholds. Precision is low for ICU and intubation despite higher recall. Death SMOTEENN used a capped training set and is not directly comparable with uncapped runs.

Results, interpretation, and limits

Experiment coverage and corrected runs

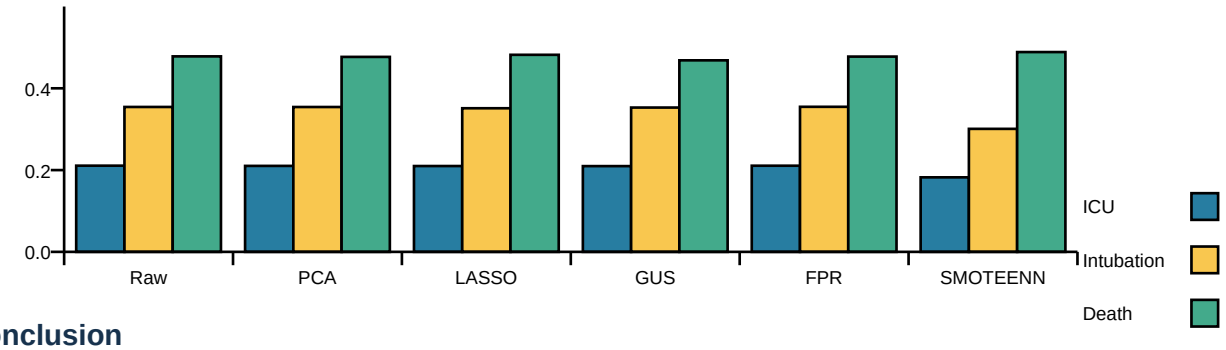
Saved method	Rows	Coverage
Raw	45	3 targets x 3 feature groups x 5 models
PCA	15	Clinical group x 3 targets x 5 models
LASSO	15	Clinical, C=1.0; 5 models for each target
SMOTEENN	42/45	100,000 training-row cap; 3 class-collapse failures
GUS / FPR	3 each	New grouped Clinical Logistic Regression runs

The corrected comparison below contains 18 grouped-split runs using Clinical features and Logistic Regression. Each target uses the same predictor-row grouping and seed across methods. SMOTEENN caps training at 100,000 rows; the test rows stay untouched.

Grouped-split F1 by feature-engineering method

Target	Raw	PCA	LASSO	GUS	FPR	SMOTEENN
ICU	0.211	0.210	0.210	0.210	0.211	0.182
INTUBED	0.354	0.354	0.351	0.353	0.355	0.301
DEATH	0.478	0.477	0.482	0.468	0.478	0.489

All plotted rows use one Logistic Regression model and one feature group. Historical raw-grid mean F1 from Demo to Clinical increased from 0.135 to 0.171 (ICU), 0.255 to 0.274 (intubation), and 0.259 to 0.455 (death); the random split allows duplicate leakage. The historical best rows on page one use that earlier split and are reported separately.



Conclusion

Historical runs show stronger ranking for death and weak positive-class precision for ICU and intubation. The data contain 812,049 exact duplicate rows, no patient identifiers, imbalanced outcomes, and predictors whose timing is not always clear. Grouped results compare feature methods for one model and feature group only. A full grouped model grid and external validation are needed before stronger claims. These models must not guide care or resource allocation.

References: Zhan, X. and Li, Y., “CT-based Patient Triage of COVID-19: Radiomics Prediction of ICU Admission, Mechanical Ventilation, and Death of Patients,” CS229 Final Report (assigned research reference). UE24CS352A, Mini-Project Assignment Guidelines, supplied course handout.